

The Guide to Decision Tree-based Algorithms in Machine Learning

+ Real Examples

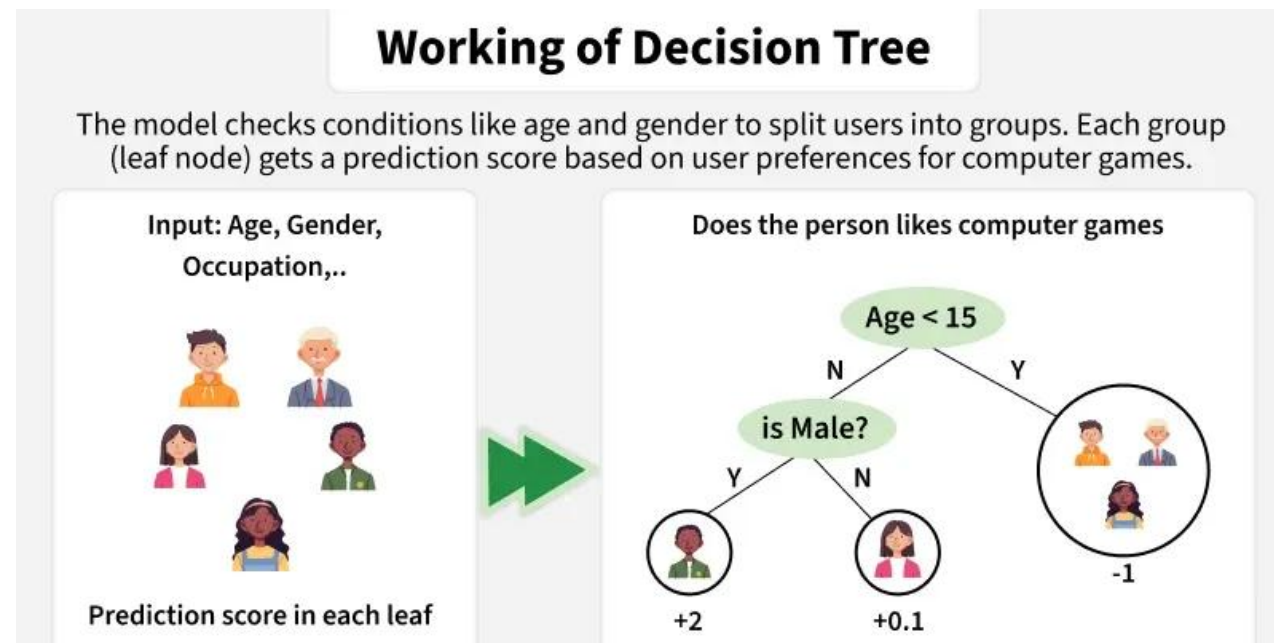


TREE-BASED MODELS (THE VISUAL LOGIC)

Making Decisions with
Flowcharts

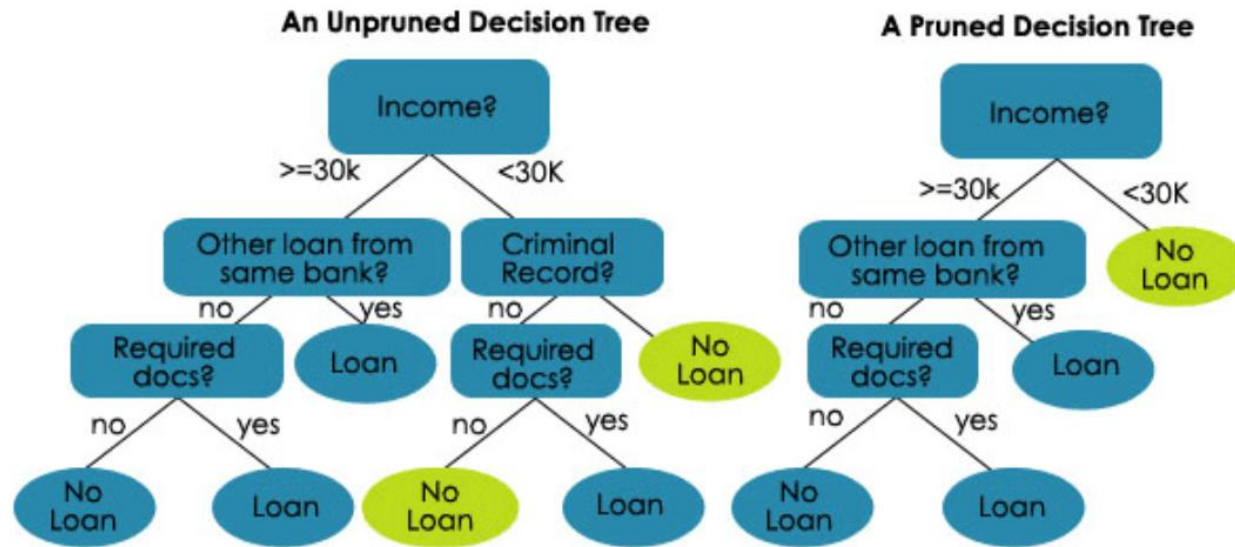
Instructor: Zaryab Ahmad
Khan

DECISION TREES



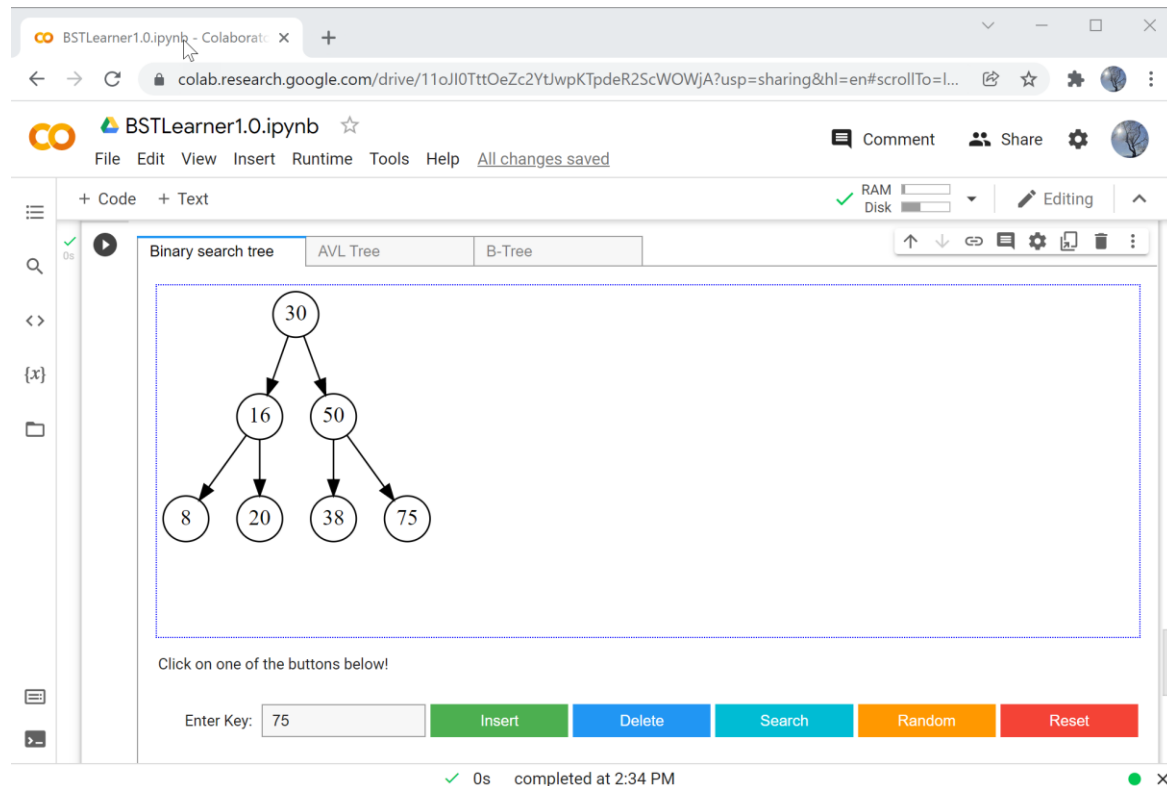
- **The Concept:** Imagine a game of "20 Questions."
- **How it works:** The algorithm asks a series of If/Else questions to split the data into categories.
- **Example:** "Is the patient's age > 30%" → If Yes, "Is their BMI > 25%" → If Yes, "Classify as High Risk."

TREE PRUNING (AVOIDING THE OVERFIT)



- **The Problem:** If a tree is allowed to grow too deep, it will create a "branch" for every single outlier in the data, leading to **Overfitting**.
 - **The Solution:** Pruning.
 - **Method:** We set a `max_depth` (e.g., limit the tree to 3 levels) to keep the logic simple and generalized for the real world.
-

VISUALIZING THE TREE



- **Transparency:** One of the biggest advantages of Decision Trees is that they aren't "Black Boxes."
- **Interpretation:** Using tools in Scikit-Learn, we can export the tree as an image, allowing humans to verify and trust the algorithm's logic.